Unit: 6 Unit 2: Exponents Radicals
Topic: Radicals

Name: ________________
Date: __________

Quiz (Habanero)

Q1. Simplify $\sqrt{64}$
(A) 8
(B) 16
(C) 32
(D) 4
(E) 6

Q2. Convert $\sqrt[3]{27}$ to rational exponent form
(A) $3^{1/2}$
(B) $3^{1/3}$
(C) $3^{2/3}$
(D) $3^{3/2}$
(E) $3^{3/3}$

Q3. What is $\sqrt{125}$ simplified?
(A) $5\sqrt{5}$
(B) $10\sqrt{5}$
(C) $15\sqrt{5}$
(D) $20\sqrt{5}$
(E) $25\sqrt{5}$

Q4. A weather station measures a $\sqrt{121}$ mm increase in rainfall. How many mm is this?
(A) 10
(B) 11
(C) 12
(D) 13
(E) 14

Q5. A biologist studies the growth of plants in an ecosystem. She finds that the growth rate is $\sqrt[3]{216}$. What is this in rational exponent form?
(A) $6^{1/2}$
(B) $6^{1/3}$
(C) $6^{2/3}$
(D) $6^{3/2}$
(E) $6^{3/3}$

Q6. Simplify $\sqrt{24}$
(A) $2\sqrt{6}$
(B) $3\sqrt{6}$
(C) $4\sqrt{6}$
(D) $5\sqrt{6}$
(E) $6\sqrt{6}$

Q7. What is $\sqrt{100}$ in simplest form?
(A) 5
(B) 10
(C) 15
(D) 20
(E) 25

Q8. A weather forecast predicts a temperature change of $\sqrt[3]{1000}$. What is this in rational exponent form?
(A) $10^{1/2}$
(B) $10^{1/3}$
(C) $10^{2/3}$
(D) $10^{3/2}$
(E) $10^{3/3}$

Open Ended Questions (FRQ)

Q9. Simplify $\sqrt{150}$ and convert it to rational exponent form. Show all steps.

Q10. A researcher studies the effects of climate change on an ecosystem. The growth rate of a species is modeled by $\sqrt[3]{x}$. If x represents the amount of rainfall in mm, find the growth rate when $x = 512$. Show all calculations.

Chef's Tips

- Tip 1: Ensure students understand the concept of radicals and rational exponents
- Tip 2: Encourage students to simplify radicals and convert between forms
- Tip 3: Remind students to show all steps in free response questions